Chapter 6 Ratio Estimators

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The ratio estimator

For the ith unit, we collect a pair of observations (x_i, y_i)

- x_i : an auxiliary variate
- The population parameter to be estimated is the ratio

$$R = \frac{\sum_{i=1}^{N} y_i}{\sum_{i=1}^{N} x_i} = \frac{Y}{X} = \frac{\bar{Y}}{\bar{X}}$$

- With a simple random sample, the corresponding sample estimate is

$$\hat{R} = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i} = \frac{\bar{y}}{\bar{x}}$$

- Aim: to obtain increased precision by taking advantages of the correlation between y_i and x_i.
- In practice, $x_i \rightarrow$ the value of y_i at some previous time in a complete census.
- The population total X of x_i must be known.
- The ratio estimate of the population total Y is

$$\hat{Y}_R = \hat{R}X = \frac{y}{x}X = \frac{\bar{y}}{\bar{x}}X$$

- The ratio estimate of the population mean $\bar{Y}$ is

$$\hat{\bar{Y}}_R = \hat{R}\bar{X} = \frac{y}{x}\bar{X} = \frac{\bar{y}}{\bar{x}}\bar{X}$$



Example

Table 6.1 shows the number of inhabitants (in 1000's) in each of a simple random sample of 49 cities drawn from the population of 196 large cities discussed in section 2.15. The problem is to estimate the total number of inhabitants in the 196 cities in 1930. The true 1920 total, X, is assumed to be known. Its value is 22,919.

Analysis: The example is a suitable one for the ratio estimate. The majority of the cities in the sample show an increase in size from 1920 to 1930 of the order of 20%.

TABLE 6.1

SIZES OF 49 LARGE UNITED STATES CITIES (in 1000's) IN 1920 (x_i) AND 1930 (y_i)

x_i	y_i	x_i	y_i	x_i	y_i
76	80	2	50	243	291
138	143	507	634	87	105
67	67	179	260	30	111
29	50	121	113	71	79
381	464	50	64	256	288
23	48	44	58	43	61
37	63	77	89	25	57
120	115	64	63	94	85
61	69	64	77	43	50
387	459	56	142	298	317
93	104	40	60	36	46
172	183	40	64	161	232
78	106	38	52	74	93
66	86	136	139	45	53
60	57	116	130	36	54
46	65	46	53	50	58
				48	75

$$y = \sum_{i=1}^{n} y_i = 6262 \qquad x = \sum_{i=1}^{n} x_i = 5054$$

- The ratio estimate of the 1930 total for all 196 cities is

$$\hat{Y}_R = \hat{R}X = \frac{y}{x}X = \frac{6262}{5054} \times 22919 = 28397$$

- The estimate based on the sample mean per city is

$$\hat{Y} = N\overline{y} = 196 \times \frac{6262}{49} = 25048$$

- The true total in 1930 is 29,351.

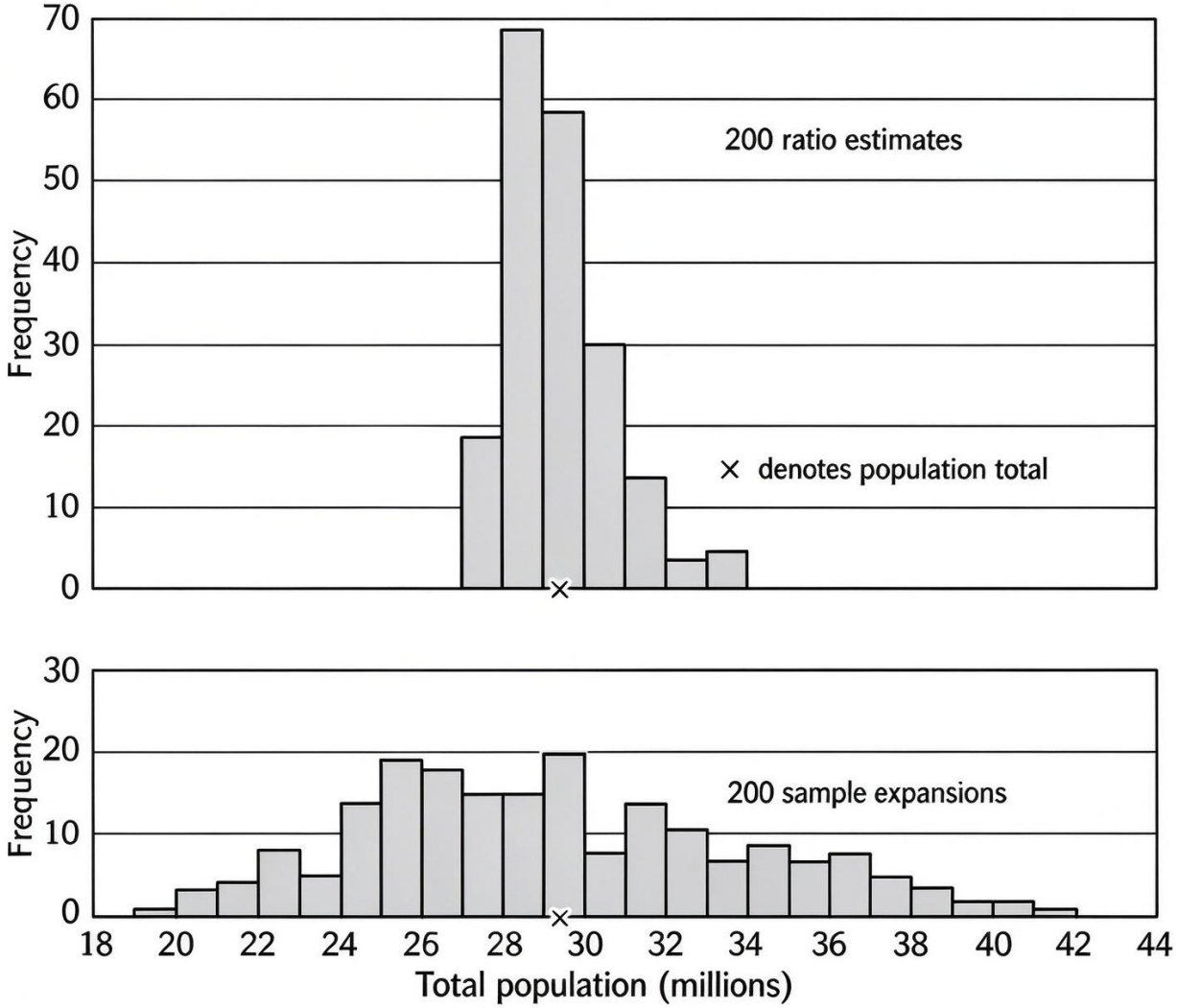


Fig. 6.1 Experimental comparison of the ratio estimate with the estimate based on the sample mean

Approximate variance of the ratio estimate

- Moderate sample size: The distribution of the ratio estimate shows a tendency to positive skewness in the kinds of populations for which the method is most often used.
- Large sample size: (a) the ratio is nearly normally distributed, (b) the large-sample formula for its variance is valid.
- Working rule to use large-sample results: If the sample size exceeds 30 and is also large enough so that the coefficients of variation of $\bar{x}$ and $\bar{y}$ are both less than 10%.

Theorem 6.1 The ratio estimates of the population ratio Y/X, the population mean $\bar{Y}$, and the population total Y are, respectively,

$$\hat{R} = \frac{\bar{y}}{\bar{x}}, \qquad \hat{\bar{Y}}_R = \hat{R}\bar{X} = \frac{\bar{y}}{\bar{x}}\bar{X}, \qquad \hat{Y}_R = \hat{R}X = \frac{\bar{y}}{\bar{x}}X.$$

In a simple random sample of size n (n large)

$$V(\hat{R}) = \frac{1-f}{n\bar{X}^2}\frac{\sum_{i=1}^{N}(y_i - Rx_i)^2}{N-1}$$

$$V(\hat{\bar{Y}}_R) = \frac{1-f}{n}\frac{\sum_{i=1}^{N}(y_i - Rx_i)^2}{N-1}$$

$$V(\hat{Y}_R) = \frac{N^2(1-f)}{n}\frac{\sum_{i=1}^{N}(y_i - Rx_i)^2}{N-1}$$

Corollary 1. There are various alternative forms of the result.

Since $\bar{Y} = R\bar{X}$,

$$V(\hat{Y}_R) = \frac{N^2(1-f)}{n(N-1)} \sum_{i=1}^{N} \left[(y_i - \bar{Y}) - R(x_i - \bar{X}) \right]^2$$

$$= \frac{N^2(1-f)}{n(N-1)} \left[\sum_{i=1}^{N} (y_i - \bar{Y})^2 + R^2 \sum_{i=1}^{N} (x_i - \bar{X})^2 - 2R \sum_{i=1}^{N} (y_i - \bar{Y})(x_i - \bar{X}) \right]$$

This can be written as

$$V(\hat{Y}_R) = \frac{N^2(1-f)}{n} \left(S_y^2 + R^2 S_x^2 - 2RS_{yx} \right)$$

$$= \frac{N^2(1-f)}{n} \left(S_y^2 + R^2 S_x^2 - 2R\rho S_y S_x \right)$$

with the correlation coefficient

$$\rho = \frac{E(y_i - \bar{Y})(x_i - \bar{X})}{\sqrt{E(y_i - \bar{Y})^2 E(x_i - \bar{X})^2}} = \frac{S_{yx}}{S_y S_x}$$

Estimation of the variance from a sample

- $\dfrac{\sum_{i=1}^{n}(y_i-\hat{R}x_i)^2}{n-1} \Rightarrow \dfrac{\sum_{i=1}^{N}(y_i-Rx_i)^2}{N-1}$ A bias of order $1/n$

- For the estimated variance $v(\hat{Y}_R)$:

$$v(\hat{Y}_R)=\frac{N^2(1-f)}{n}\frac{\sum_{i=1}^{n}(y_i-\hat{R}x_i)^2}{n-1}$$

- Another expression:

$$v(\hat{Y}_R)=\frac{N^2(1-f)}{n}\frac{\left(\sum_{i=1}^{n}y_i^2+\hat{R}^2\sum_{i=1}^{n}x_i^2-2\hat{R}\sum_{i=1}^{n}y_ix_i\right)}{n-1}$$

$$=\frac{N^2(1-f)}{n}(s_y^2+\hat{R}^2s_x^2-2\hat{R}s_{yx})$$

where $s_{yx}=\sum_{i=1}^{n}(y_i-\bar{y})(x_i-\bar{x})/(n-1)$ is the sample covariance.

- Two forms for the sample estimate of the variance for $\hat{R}$

$$v_1(\hat{R}) = \frac{1-f}{n\bar{X}^2}\frac{\sum_{i=1}^{n}(y_i - \hat{R}x_i)^2}{n-1} = \frac{1-f}{n\bar{X}^2}(s_y^2 + \hat{R}^2 s_x^2 - 2\hat{R}s_{yx})$$

$$v_2(\hat{R}) = \frac{1-f}{n\bar{x}^2}\frac{\sum_{i=1}^{n}(y_i - \hat{R}x_i)^2}{n-1} = \frac{1-f}{n\bar{x}^2}(s_y^2 + \hat{R}^2 s_x^2 - 2\hat{R}s_{yx})$$

- Based on $v_2(\hat{R})$, we can take

$$v_2(\hat{Y}_R) = X^2 v_2(\hat{R})$$

- Hard to say which one of the above two estimate is better.



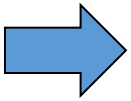
Example

There are 56 companies, whose total profits in last year were 86436 thousand dollars. In order to estimate their profits in this year, a simple random sample composed of 12 companies were selected. Their profits in the two years, denoted by x_i, y_i, are listed in the following table. Please estimate the **total profits** for all the 56 companies in this year and the s.e.

	1	2	3	4	5	6
x_i	764	1642	957	1324	2131	1176
y_i	853	1835	1028	1512	2286	1354
	7	8	9	10	11	12
x_i	1618	1532	834	1432	1728	1214
y_i	1826	1721	958	1648	1904	1534

$$\sum_{i=1}^{12} x_i = 16352,\ \sum_{i=1}^{12} x_i^2 = 24043730,\ \bar{x} = 1362.67,$$

$$\sum_{i=1}^{12} y_i = 18459,\ \sum_{i=1}^{12} y_i^2 = 30405031,\ \bar{y} = 1538.25,\ \sum_{i=1}^{12} x_i y_i = 27016552$$



$$\hat{Y}_R = \hat{R}X = \frac{18459}{16352} \times 86436 = 97573.52$$

$$v(\hat{Y}_R) = \frac{N^2(1-f)}{n} \frac{\sum_{i=1}^{n}(y_i - \hat{R}x_i)^2}{n-1}$$

$$= \frac{56^2 \times (1 - \frac{12}{56})}{12 \times 11} \left(\sum_{i=1}^{12} y_i^2 + \hat{R}^2 \sum_{i=1}^{12} x_i^2 - 2\hat{R} \sum_{i=1}^{12} y_i x_i \right) = 909860$$

$$se(\hat{Y}_R) = \sqrt{v(\hat{Y}_R)} = 953.87$$

Confidence limits

- If the sample is large enough so that the normal approximation applies, confidence limits for Y and R are:

$$Y: \quad \hat{Y}_R \pm z_{\alpha/2}\sqrt{v(\hat{Y}_R)} \qquad \bar{Y}: \quad \hat{\bar{Y}}_R \pm z_{\alpha/2}\sqrt{v(\hat{\bar{Y}}_R)}$$

$$R: \quad \hat{R} \pm z_{\alpha/2}\sqrt{v(\hat{R})}$$

- Condition: $n \geq 30$ and n is large enough so that the

$$\text{cv}(\bar{y}) < 0.1, \text{cv}(\bar{x}) < 0.1.$$

- If these conditions DO NOT apply: (a) the formula for $v(\hat{R})$ tends to give values that are too low; (b) the positive skewness in the distribution of $\hat{R}$ may become noticeable.

Comparison of the ratio estimate with the mean per unit

- The type of estimate of Y that was studied in preceding chapters is $N\bar{y}$, where $\bar{y}$ is the mean per unit for the sample (in simple random sampling) or a weighted mean per unit (in stratified random sampling).
- Estimates of this kind are called estimates based on the mean per unit or estimates obtained by simple expansion.

Theorem 6.2 In large samples, with simple random sampling, the ratio estimate $\hat{Y}_R$ has a smaller variance than the estimate $\hat{Y} = N\bar{y}$ obtained by simple expansion, if

$$\rho > \frac{1}{2}\left(\frac{S_x}{\bar{X}}\right)\Big/\left(\frac{S_y}{\bar{Y}}\right) = \frac{1}{2}\frac{\text{coefficient of variation of } x_i}{\text{coefficient of variation of } y_i}$$

Proof: Compare the two variances

$$V(\hat{Y}) = \frac{N^2(1-f)}{n}S_y^2$$

$$V(\hat{Y}_R) = \frac{N^2(1-f)}{n}(S_y^2 + R^2S_x^2 - 2R\rho S_y S_x)$$



Example (Continued)

$$\hat{Y}_R = \hat{R}X = \frac{18459}{16352} \times 86436 = 97573.52 \qquad se(\hat{Y}_R) = \sqrt{v(\hat{Y}_R)} = 953.87$$

$$\hat{Y} = N\bar{y} = 56 \times \frac{18459}{12} = 86142 \qquad se(\hat{Y}) = \sqrt{v(\hat{Y})} = 6126.08$$

Here

$$cv_x = \frac{s_x}{\bar{x}} = 0.2937 \qquad cv_y = \frac{s_y}{\bar{y}} = 0.2779 \qquad r = 0.99$$

Ratio estimates in stratified random sampling

A ratio estimate of the population total Y: two ways

- One is to make a separate ratio estimate of the total of each stratum and add these totals.
- y_h, x_h : the sample totals in the hth stratum
- X_h : the stratum total of the x_{hi}
- The separate ratio estimate:

$$\hat{Y}_{Rs} = \sum_{h=1}^{L} \hat{Y}_{Rh} = \sum_{h=1}^{L} \hat{R}_h X_h = \sum_{h=1}^{L} \frac{y_h}{x_h} X_h = \sum_{h=1}^{L} \frac{\bar{y}_h}{\bar{x}_h} X_h$$

$$\bar{y}_{Rs} = \sum_{h=1}^{L} W_h \bar{y}_{Rh} = \sum_{h=1}^{L} W_h \hat{R}_h \bar{X}_h = \sum_{h=1}^{L} W_h \frac{\bar{y}_h}{\bar{x}_h} \bar{X}_h = \frac{\hat{Y}_{Rs}}{N}$$

$$\hat{Y}_{Rs} = \sum_{h=1}^{L} \hat{Y}_{Rh} = \sum_{h=1}^{L} \hat{R}_h X_h = \sum_{h=1}^{L} \frac{y_h}{x_h} X_h = \sum_{h=1}^{L} \frac{\bar{y}_h}{\bar{x}_h} X_h$$

- No assumption is made that the true ratio remains constant from stratum to stratum.
- Requires the separate totals and means X_h, $\bar{X}_h$

Theorem 6.4 If an independent simple random sample is drawn in each stratum and sample sizes are large in all strata,

$$V(\hat{Y}_{Rs}) \approx \sum_{h=1}^{L} \frac{N_h^2(1-f_h)}{n_h}(S_{yh}^2 + R_h^2 S_{xh}^2 - 2R_h \rho_h S_{yh} S_{xh})$$

where $R_h = Y_h / X_h$ is the true ratio in stratum h, and ρ_h is defined as before in each stratum.

The combined ratio estimate

- An alternative estimate : a single combined ratio
- The standard estimates of the population totals Y and X:

$$\hat{Y}_{st} = \sum_{h=1}^{L} N_h \bar{y}_h, \qquad \hat{X}_{st} = \sum_{h=1}^{L} N_h \bar{x}_h$$

- The combined ratio estimate:

$$\hat{Y}_{Rc} = \hat{R}_c X = \frac{\hat{Y}_{st}}{\hat{X}_{st}} X = \frac{\bar{y}_{st}}{\bar{x}_{st}} X$$

where $\bar{y}_{st} = \hat{Y}_{st} / N$, $\bar{x}_{st} = \hat{X}_{st} / N$ are the estimated means.

Theorem 6.5 If the total sample size n is large,

$$V(\hat{Y}_{Rc}) \approx \sum_{h=1}^{L} \frac{N_h^2 (1 - f_h)}{n_h} (S_{yh}^2 + R^2 S_{xh}^2 - 2R\rho_h S_{yh} S_{xh})$$

Comparison of the combined and separate estimates

$$V(\hat{Y}_{Rs}) \approx \sum_{h=1}^{L} \frac{N_h^2(1-f_h)}{n_h}(S_{yh}^2 + R_h^2 S_{xh}^2 - 2R_h\rho_h S_{yh}S_{xh})$$

$$V(\hat{Y}_{Rc}) \approx \sum_{h=1}^{L} \frac{N_h^2(1-f_h)}{n_h}(S_{yh}^2 + R^2 S_{xh}^2 - 2R\rho_h S_{yh}S_{xh})$$



$$V(\hat{Y}_{Rc}) - V(\hat{Y}_{Rs})$$

$$\approx \sum_{h=1}^{L} \frac{N_h^2(1-f_h)}{n_h}(R^2 S_{xh}^2 - R_h^2 S_{xh}^2 - 2R\rho_h S_{yh}S_{xh} + 2R_h\rho_h S_{yh}S_{xh})$$

$$= \sum_{h=1}^{L} \frac{N_h^2(1-f_h)}{n_h}(R^2 S_{xh}^2 - 2RR_h S_{xh}^2 + R_h^2 S_{xh}^2$$

$$+ 2RR_h S_{xh}^2 - 2R_h^2 S_{xh}^2 - 2R\rho_h S_{yh}S_{xh} + 2R_h\rho_h S_{yh}S_{xh})$$

$$= \sum_{h=1}^{L} \frac{N_h^2(1-f_h)}{n_h}[(R-R_h)^2 S_{xh}^2 + 2(R_h - R)(\rho_h S_{yh}S_{xh} - R_h S_{xh}^2)]$$

$$V(\hat{Y}_{Rc}) - V(\hat{Y}_{Rs}) = \sum_{h=1}^{L} \frac{N_h^2(1-f_h)}{n_h}[(R-R_h)^2 S_{xh}^2 + 2(R_h - R)(\rho_h S_{yh} S_{xh} - R_h S_{xh}^2)]$$

- If the ratio estimate is appropriate, the last term on the right is usually small.

$$y_{hi} = R_h x_{hi} + e_{hi} \quad \rightarrow \quad \mathrm{cov}(y_{hi}, x_{hi}) = \mathrm{cov}(R_h x_{hi}, x_{hi}) = R_h S_{xh}^2 = \rho_h S_{yh} S_{xh}$$

- Unless $R_h = R\ (h = 1, ..., L)$, the use of a separate ratio estimate is likely to be more precise, given the sample in each stratum is large enough.
- With only a small sample in each stratum, the combined estimate is to be recommended.



Example

The data come from a census of all farms in Jefferson County, Iowa. In this example y_{hi} represents acres in corn and x_{hi} acres in the farm. The population is divided into two strata, the first stratum containing farms of as many as 160 acres.

We assume a sample of 100 farms. When stratified sampling is used, we will suppose that 70 farms are taken from stratum 1 and 30 from stratum 2, this being roughly the optimum allocation.

The data are given in Table 6.3. We consider five methods of estimating the population mean corn acres per farm. The fpc are ignored.

TABLE 6.3

DATA FROM JEFFERSON COUNTY, IOWA

Strata	Size (farm acres)	N_h	S_{yh}^2	S_{yxh}	S_{yh}^2	R_h
1	0–160	1580	312	494	2055	0.2350
2	More than 160	430	922	858	7357	0.2109
For complete pop.		2010	620	1453	7619	0.2242

Strata	$\bar{Y}_h$	$\bar{X}_h$	n_h	$Q_h = W_h^2/n_h$	V_h'	V_h''
1	19.40	82.56	70	0.008828	193	194
2	51.63	244.85	30	0.001525	887	907
For complete pop.	26.30	117.28	100			

1. Simple random sample: mean per farm estimate.

$$V_1 = V(\bar{y}) = \frac{S_y^2}{n} = \frac{620}{100} = 6.2$$

2. Simple random sample: ratio estimate.

$$V_2 = V(\bar{y}_R) = \frac{1}{n}(S_y^2 + R^2 S_x^2 - 2RS_{yx})$$

$$= \frac{1}{100}(620 + 0.2242^2 \times 7619 - 2 \times 0.2242 \times 1453) = 3.51$$

3. Stratified random sample: mean per farm estimate.

$$V_3 = V(\bar{y}_{st}) = \sum_{h=1}^{L} \frac{W_h^2 S_{y_h}^2}{n_h} = 4.16$$

4. Stratified random sample: ratio estimate using a separate ratio

$$V_4 = V(\hat{\bar{Y}}_{Rs}) \approx \sum_{h=1}^{L} \frac{W_h^2}{n_h}(S_{yh}^2 + R_h^2 S_{xh}^2 - 2R_h S_{yxh}) = 3.06$$

5. Stratified random sample: ratio estimate using a combined ratio

$$V_5 = V(\hat{\bar{Y}}_{Rc}) \approx \sum_{h=1}^{L} \frac{W_h^2}{n_h}(S_{yh}^2 + R^2 S_{xh}^2 - 2RS_{yxh}) = 3.10$$

Sampling method	Method of Estimation	Relative Precision
1. Simple random	Mean per farm	100
2. Simple random	Ratio	177
3. Stratified random	Mean per farm	149
4. Stratified random	Separate ratio	203
5. Stratified random	Combined ratio	200

- Stratification by size of farm has the same general purpose as a ratio estimate in which the denominator is farm size.
- Both devices diminish the effect of variations in farm size on the sampling error of the estimated mean corn acres per farm.

- Relevant points: between introducing a factor into the stratification or utilizing it in the method of estimation

  (a) some factors (e.g. geographical location) are more easily introduced into the stratification;

  (b) it depends on the nature of the relation between y_i and x_i ;

  (c) if some important variates are roughly proportional to x_i , and others are proportional to z_i , it is better to use ratio estimates.

Multivariate ratio estimates

- If there are p auxiliary x-variables $(x_1, x_2, ..., x_p)$ are available, we can construct multivariate ratio $\hat{Y}_{MR}$:

$$\hat{Y}_{MR} = W_1\hat{Y}_{R_1} + W_2\hat{Y}_{R_2} + \cdots + W_p\hat{Y}_{R_p}$$

$$= W_1\frac{\bar{y}}{\bar{x}_1}X_1 + W_2\frac{\bar{y}}{\bar{x}_2}X_2 + \cdots + W_p\frac{\bar{y}}{\bar{x}_p}X_p$$

where W_i are weights to be determined to maximize the precision of $\hat{Y}_{MR}$ subject to $\sum W_i = 1$.

- This type of estimate appears appropriate when the regression of y on $x_1, x_2, ..., x_p$ is linear and passes through the origin.
- The population totals X_i must be known.

- Consider two x-variates:

$$\hat{Y}_{MR} - Y = W_1\hat{Y}_{R_1} + W_2\hat{Y}_{R_2} - W_1Y - W_2Y = W_1(\hat{Y}_{R_1} - Y) + W_2(\hat{Y}_{R_2} - Y)$$

Assuming negligible bias

$$V(\hat{Y}_{MR}) = W_1^2V(\hat{Y}_{R_1}) + 2W_1W_2\,\mathrm{cov}(\hat{Y}_{R_1}, \hat{Y}_{R_2}) + W_2^2V(\hat{Y}_{R_2})$$
$$= W_1^2V_{11} + 2W_1W_2V_{12} + W_2^2V_{22}$$

where $V_{11} = V(\hat{Y}_{R_1})$, $V_{22} = V(\hat{Y}_{R_2})$ and $V_{12} = \mathrm{cov}(\hat{Y}_{R_1}, \hat{Y}_{R_2})$.

- With $W_1 + W_2 = 1$, minimizing the variance yields

$$W_1 = \frac{V_{22} - V_{12}}{V_{11} + V_{22} - 2V_{12}}, \quad W_2 = \frac{V_{11} - V_{12}}{V_{11} + V_{22} - 2V_{12}}$$

$$V_{\min}(\hat{Y}_{MR}) = \frac{V_{11}V_{22} - V_{12}^2}{V_{11} + V_{22} - 2V_{12}}$$

- In practice, the weights are determined from estimated variances and covariances v_{ij} .

$$v_{11} = \frac{N^2(1-f)}{n}(s_y^2 + \hat{R}_1^2 s_{x_1}^2 - 2\hat{R}_1 s_{yx_1}) = \frac{(1-f)\hat{Y}^2}{n}(c_{yy} + c_{11} - 2c_{y1})$$

$$v_{22} = \frac{N^2(1-f)}{n}(s_y^2 + \hat{R}_2^2 s_{x_2}^2 - 2\hat{R}_2 s_{yx_2}) = \frac{(1-f)\hat{Y}^2}{n}(c_{yy} + c_{22} - 2c_{y2})$$

where $c_{yy} = s_y^2 / \bar{y}^2$ etc.

$$v_{12} = \frac{N^2(1-f)}{n}(s_y^2 + \hat{R}_1\hat{R}_2 s_{x_1x_2} - \hat{R}_1 s_{yx_1} - \hat{R}_2 s_{yx_2}) = \frac{(1-f)\hat{Y}^2}{n}(c_{yy} + c_{12} - c_{y1} - c_{y2})$$

- A useful matrix:

$$C = \begin{pmatrix} c_{yy} & c_{y1} & c_{y2} \\ c_{y1} & c_{11} & c_{12} \\ c_{y2} & c_{12} & c_{22} \end{pmatrix}$$

$$v_{11} = v(\hat{Y}_{R1}) = \frac{N^2(1-f)}{n}\frac{\sum_{i=1}^{n}(y_i - \hat{R}_1 x_{1i})^2}{n-1}$$

$$= \frac{N^2(1-f)}{n}(s_y^2 + \hat{R}_1^2 s_{x_1}^2 - 2\hat{R}_1 s_{yx_1}) = \frac{(1-f)\hat{Y}^2}{n}(c_{yy} + c_{11} - 2c_{y1})$$

$$v_{22} = v(\hat{Y}_{R2}) = \frac{N^2(1-f)}{n}\frac{\sum_{i=1}^{n}(y_i - \hat{R}_2 x_{2i})^2}{n-1}$$

$$= \frac{N^2(1-f)}{n}(s_y^2 + \hat{R}_2^2 s_{x_2}^2 - 2\hat{R}_2 s_{yx_2}) = \frac{(1-f)\hat{Y}^2}{n}(c_{yy} + c_{22} - 2c_{y2})$$

$$v_{12} = \mathrm{cov}(\hat{Y}_{R1}, \hat{Y}_{R2}) = \frac{N^2(1-f)}{n}\frac{\sum_{i=1}^{n}(y_i - \hat{R}_1 x_{1i})(y_i - \hat{R}_2 x_{2i})}{n-1}$$

$$= \frac{N^2(1-f)}{n}\frac{\sum_{i=1}^{n}(y_i - \bar{y} - \hat{R}_1 x_{1i} + \hat{R}_1 \bar{x}_1)(y_i - \bar{y} - \hat{R}_2 x_{2i} + \hat{R}_2 \bar{x}_2)}{n-1}$$

$$= \frac{N^2(1-f)}{n}(s_y^2 + \hat{R}_1 \hat{R}_x s_{x_1 x_2} - \hat{R}_1 s_{yx_1} - \hat{R}_2 s_{yx_2}) = \frac{(1-f)\hat{Y}^2}{n}(c_{yy} + c_{12} - c_{y1} - c_{y2})$$



Example

We are interested in estimation of cotton production in a district, which is composed of 301 villages. A SRS of 18 villages were selected. The obtained data include the cotton production y_i, the acreage x_{1i} and the proportion of good seedling x_{2i}. The total cotton acreage in the district is $X_1 = 7450$ square kilometers, and the average proportion of good seedling is $\overline{X}_2 = 40.10\%$. There are some results:

$$\overline{y} = 13.7967,\ \overline{x}_1 = 24.4389,\ \overline{x}_2 = 38.4444,\ s_y^2 = 35.4958$$

$$s_{x_1}^2 = 74.6789,\ s_{x_2}^2 = 174.9671,\ s_{yx_1} = 42.2617,\ s_{yx_2} = 46.5118,\ s_{x_1x_2} = 114.3081$$

And $\frac{1-f}{n} = 0.0522332$. Estimate the total cotton production by the ratio estimation with the acreage and the proportion of good seedling as auxiliary variates.

- Ratio estimate with only one auxiliary variate:

$$\hat{Y}_{R_1} = \hat{R}_1 X_1 = \frac{13.7967}{24.43899} \times 7450 = 4205.80$$

$$v(\hat{Y}_{R_1}) = \frac{N^2(1-f)}{n}(s_y^2 + \hat{R}_1^2 s_{x_1}^2 - 2\hat{R}_1 s_{yx_1}) = 56158.11$$

$$se(\hat{Y}_{R_1}) = 236.98$$

$$\hat{Y}_{R_2} = \hat{R}_2 X_2 = N\hat{R}_2 \bar{X}_2 = 301 \times \frac{13.7967}{38.4444} \times 40.10 = 4331.63$$

$$v(\hat{Y}_{R_2}) = \frac{N^2(1-f)}{n}(s_y^2 + \hat{R}_2^2 s_{x_2}^2 - 2\hat{R}_2 s_{yx_2}) = 126844.51$$

$$se(\hat{Y}_{R_2}) = 356.15$$

- Ratio estimate with two auxiliary variate

$$W_1 = \frac{v_{22} - v_{12}}{v_{11} + v_{22} - 2v_{12}} = 0.6568 \qquad W_2 = 1 - W_1 = 0.3432$$

$$\hat{Y}_{MR} = W_1 \hat{Y}_{R_1} + W_2 \hat{Y}_{R_2} = 4248.98$$

$$v(\hat{Y}_{MR}) = \frac{(v_{11}v_{22} - v_{12})^2}{v_{11} + v_{22} - 2v_{12}} = 33014.94$$

$$se(\hat{Y}_{MR}) = 181.20$$

The ratio estimate with two auxiliary variate is more precise.